

Script of the Design Course

Design Course: RF- and Microwave Circuits and Systems

Prof. Dr. Rüdiger Quay,

ruediger.quay@inatech.uni-freiburg.de

19.04.2026

Version 12.0



EEH, Fritz-Hüttinger chair energy-efficient
high-frequency electronics
Freiburg University

Contents

1	Abstract	6
2	Active Modeling: Large-Signal Model of a GaAs Field Effect (Block 2: Task 8)	10
2.1	Active Modeling: Large-Signal Model of a GaAs Field Effect (Block 2: Task 8)	10
2.2	Example	15
3	Tasks 8 and 9: Large-Signal Models and Simulation (Block 2b)	17
3.1	Tasks 8 and 9: Large-Signal Models and Simulation (Block 2b)	17

Introduction

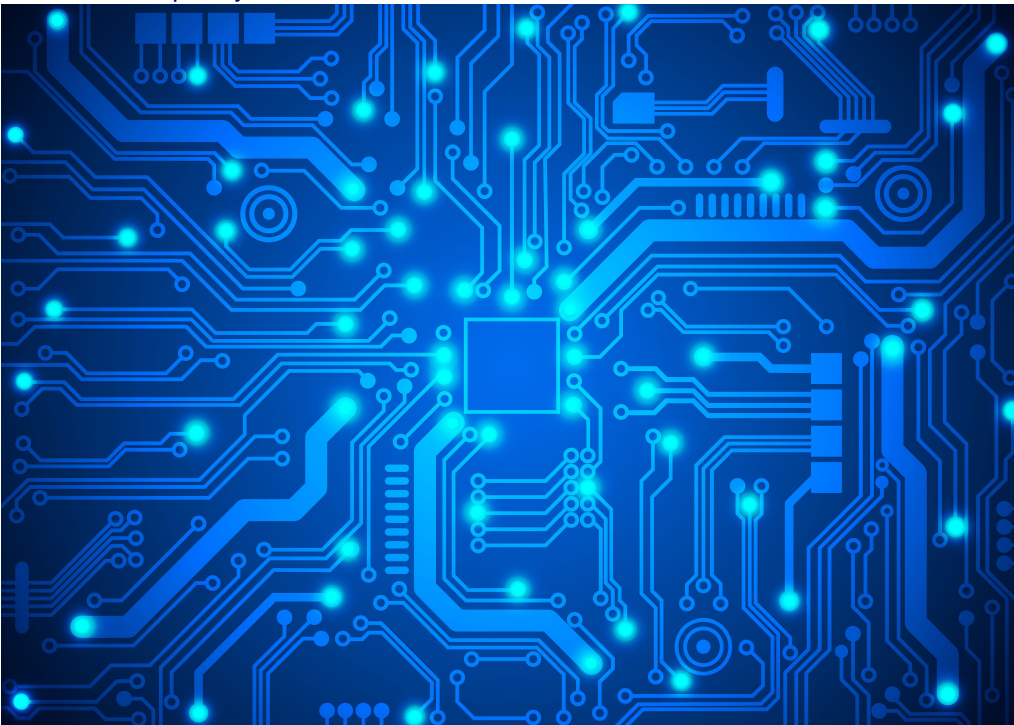


Introduction



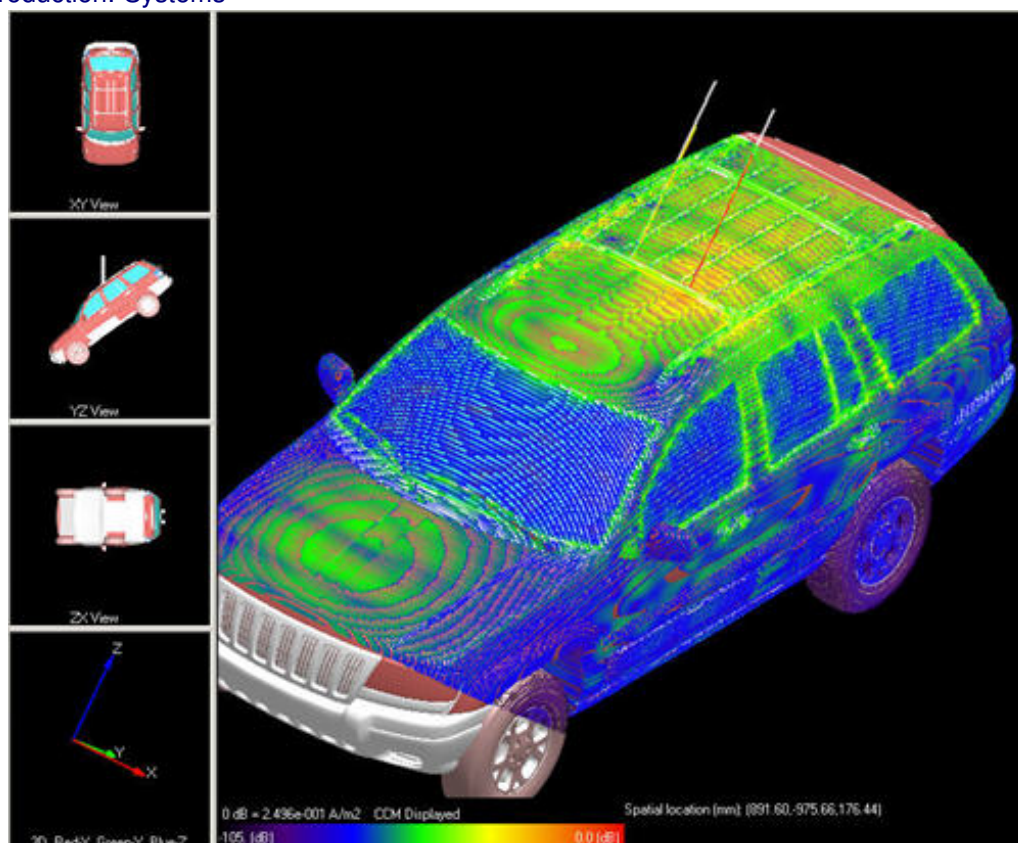
1.4

Introduction: Complexity



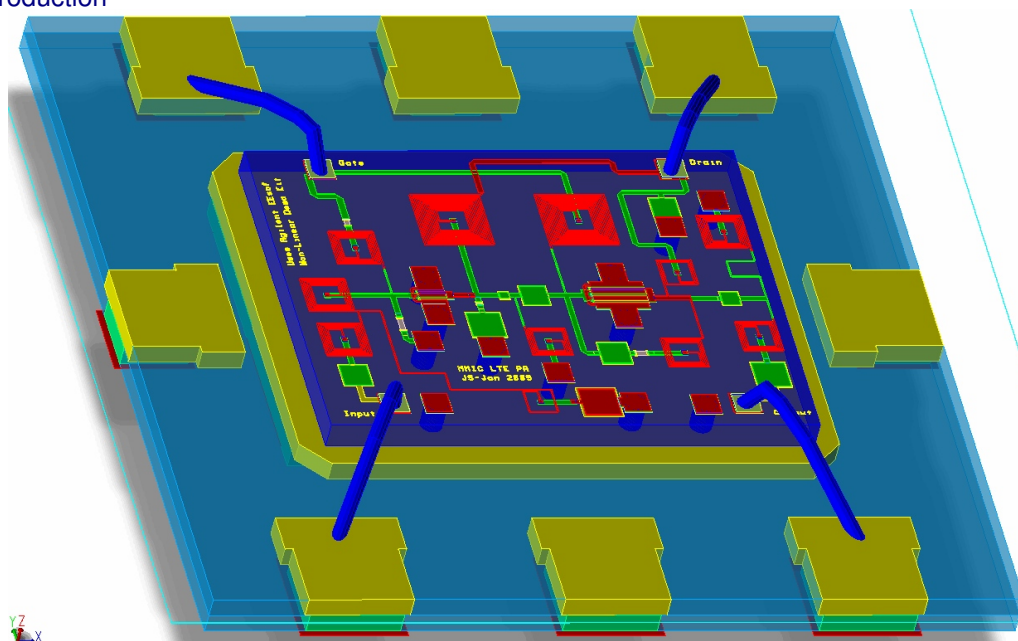
1.5

Introduction: Systems



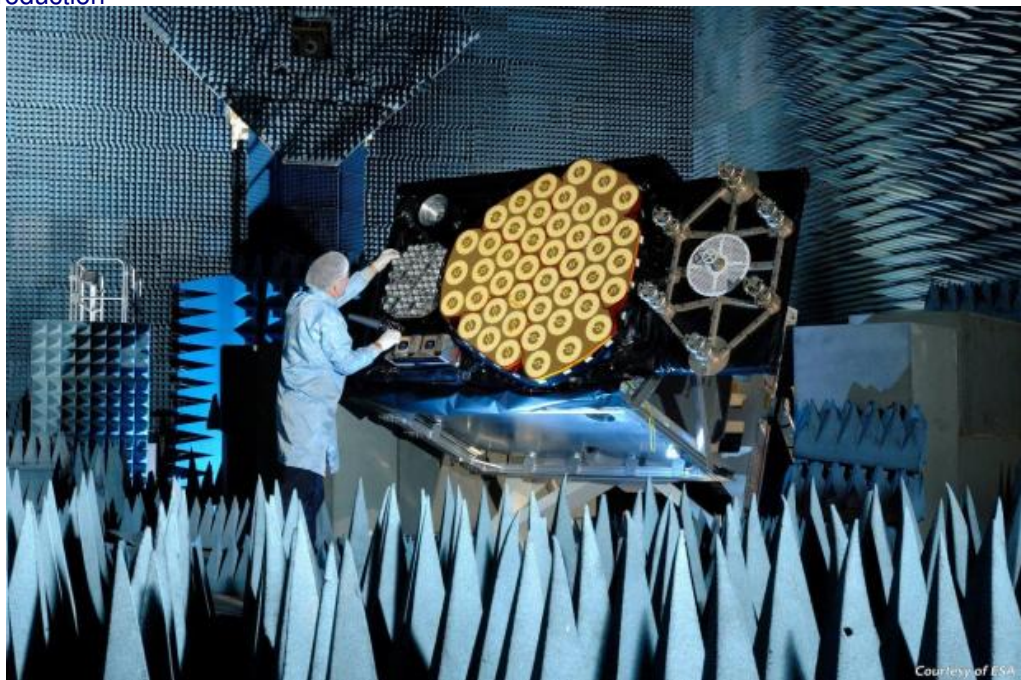
1.6

Introduction



!!!! Frequency domain !!!! vs. Cadence Virtuoso: Time domain

1.7



Dates with exercise

- Mo 1.) 20.04.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 2.) 27.04.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 3.) 4.05.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 4.) 11.05.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 5. 18.05.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 6. 25.05.2026 no lecture, pentacost break SR 00 014 (G.-Köhler-Allee 078)
- Mo 7. 1.06.2026 live SR 00 014 (G.-Köhler-Allee 078)

1.9

Dates with exercise II

- Mo 8. 8.6.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 9. 15.06.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 10. 22.06.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 11. 29.06.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 12. 6.07.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 12. 13.07.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 13. 20.07.2026 Presentation of final results from competition 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)

1.10

Date and Location

Dates with exercise

1. Mo, 20.4.2026: 16-17.00 h Introduction and group formation
2. Block 1: Mo., 27.4.2026: (issue block 1a on ILIAS)
3. Block 1: Mo., 4.5.2026: (issue block 1b on ILIAS)
4. Block 2: Mo., 11.5.2026 (issue block 2a on ILIAS), (Deadl. Block 1)
5. Block 2: Mo., 18.5.2026 (issue block 2b on ILIAS)
6. Block 3: Mo., 1.6.2026 (deadline block 2 on ILIAS), (issue block 3a on ILIAS)
7. Block 3: Mo., 08.6.2026 (issue block 3b on ILIAS)
8. Block 4: Mo., 15.6.2026 (Deadline block 3), (issue block 4a on ILIAS)
9. Block 4: Mo., 22.6.2026, (Issue Block 4b, COMPETITION !!!)
10. Block 5: Mo. 29.6.2026 (deadline block 4), (issue block 5a on ILIAS)
11. Block 5: Mo., 06.7.2026 : (issue block 5b ON ILIAS)
12. Mo., 13.7.2026, Final presentations based on 4b (deadline block 5 on ILIAS)

1.11

Rules of the Exercises

- 5 blocks Exercises (every one/two weeks) with overall max. 10 tasks in a given format...
- First: starting 27.4.2026... (two weeks to do, sometimes more)
- 4 blocks to be completed successfully (with 50% correct) (three people per group)
- Voluntary Competition: Final presentation (5 mins) of the group to the plenary based on a task
- Exercises are given out on ILIAS and also submitted there.
- **Timing is day of the deadline, 16 h**

1.12

1 Abstract

The students will be enabled to understand, design and layout modern RF- and microwave components and systems by means of the electronic design environment Advanced Design System including the two- and three dimensional electromagnetic simulators Momentum and EMPro 3D. The detailed use of a complex RF-software environment is a dedicated target of this course. This includes the numerical analysis of complex passive and active devices, the design and layout of hybrid and integrated circuits, and their packaging and signal flow. The students will be competent to design and layout passive and active RF-structures including packages and interconnects and circuits of relevance to everyday communication and sensing. The competence includes in-depth understanding and treatment of complex microwave systems and of general system design including the treatment of complex modulated signal flows.

Introduction

Materials and Script

The material is provided at the beginning and during the lecture.

Email: ruediger.quay@inatech.uni-freiburg.de

- Foils: Intro materials from Keysight
- ADS Video Library on www.youtube.com
- Script (supporting materials)
- Exercises

1.13

Videos

1. Watch ADS videos on youtube using ADS : www.youtube.com , Keyword: ADS Video Library
2. Basics and setup:
3. <https://www.youtube.com/watch?v=04BfajzXnmM&list=PLtq84kH8xZ9HNMCB7xKGXrciiEqqHNQGz>
4. Understanding Workspaces:
5. <https://www.youtube.com/watch?v=7jNm400BaiE&list=PLtq84kH8xZ9HNMCB7xKGXrciiEqqHNQGz&index=2>
6. The basics of schematics:
7. <https://www.youtube.com/watch?v=wqyyOCzxGS0&list=PLtq84kH8xZ9HNMCB7xKGXrciiEqqHNQGz&index=4>
8. Data displays:
9. <https://www.youtube.com/watch?v=jIg10RihLQk&list=PLtq84kH8xZ9HNMCB7xKGXrciiEqqHNQGz&index=6>

1.14

WWW Support 1

ILIAS Support

<https://ilias.uni-freiburg.de>

search for: Design Course: RF- and Microwave Systems

password: design2026

1.15

WWW Support 2

This exercise is recorded, attendance is, however, required



1.16

Windows Access to the Uni Network

- Get login at university, get additional TF login
- Install VPN from University server
- use ssh from linux
- If you start from Windows:
- Use, e.g., xming as windows X-server (for export of the display)
- Install Putty ssh client on windows

1.17

Form groups of Three

- The exercises are submitted from a group of three people
- Find two partners !!!
- Send your group to ruediger.quay@inatech.uni-freiburg.de, including TF login name.
- Exercises start april 27th 2026 !

1.18

Install an ADS Version on a local computer

Last Updated: 05 March 2020

ADS - Windows® Operating Systems	
Product	Windows
ADS 2020 / ADS 2020 Update 1.0 / ADS 2020 Update 2.0 64-bit Operating Systems	Windows 10 (64-bit) Windows 7 Enterprise (64-bit)
ADS 2019/ ADS 2019 Update 1.0 64-bit Operating Systems	Windows 10 (64-bit) Windows 7 Enterprise (64-bit)
ADS 2017/ ADS 2017 Update 1.0 64-bit Operating Systems	Windows 10 (64-bit) Windows 8 Enterprise (64-bit) Windows 7 Enterprise (64-bit)
ADS 2016.01 64-bit Operating Systems	Windows 10 (64-bit)* Windows 8 Enterprise (64-bit) Windows 7 Enterprise (64-bit)
ADS 2015.01 64-bit Operating Systems	Windows 8 Enterprise (64-bit) Windows 7 Enterprise (64-bit)

You can work on the aragorn but local has proven to be easier...

Organize your data in a good structure

1.19

Login to Application Server for the license

From Remote: Create RAS Access in Mylogin

Go into University net with VPN: check VPN Uni Freiburg

In case you use Windows to do that: use putty and install xming as windows X-server

Application Server: `aragorn.imtek.uni-freiburg.de`, `ssh-X-1 (yourlogin) aragorn.imtek.uni-freiburg.de`

Shells usable: `bash` and `tcsh`

Start: `ads` after using `ssh` to the server

Licence Server will be asked: `27010(at)frodo.imtek.uni-freiburg.de`

1.20

Backup Server !

Application Server: `samweis.imtek.uni-freiburg.de` (only via VPN)

1.21

Recommendation

Use Licence server only, install ADS on your local device ! Use ADS 2023 Update 1.0

1.22

The following references may be useful to provide more detailed information.

References

- List of useful ADS Videos
- Keysight Software Support: <http://www.keysight.com/>
- Introduction to ADS, ADS_Basics.pdf (ILIAS)
- Momentum Momentum_Basics.pdf (ILIAS)

1.23

Learnings: Tasks 2-5

1. Which parameters of a line are affected when changing the length ?
2. Estimate losses: Losses in dB: a difference in 1 dB ? What is doubling the power ?
3. Always look at amplitude and phase !
4. Scale of 50 dB ?
5. S-parameters of a pure inductor to 50 Ohms ? : How does it look ?
6. S11 Input return loss, S22 out return loss, S21: forward transmission, S12: reverse isolation
7. Reality of -100 dB, at going to DC ?

5.25

Task 2-5 (Block 1) continued: LineCalc

1. Power: 10 dB Factor 10, 20 dB Factor 100
2. Document your findings for all three ...

5.26

2 Active Modeling: Large-Signal Model of a GaAs Field Effect (Block 2: Task 8)

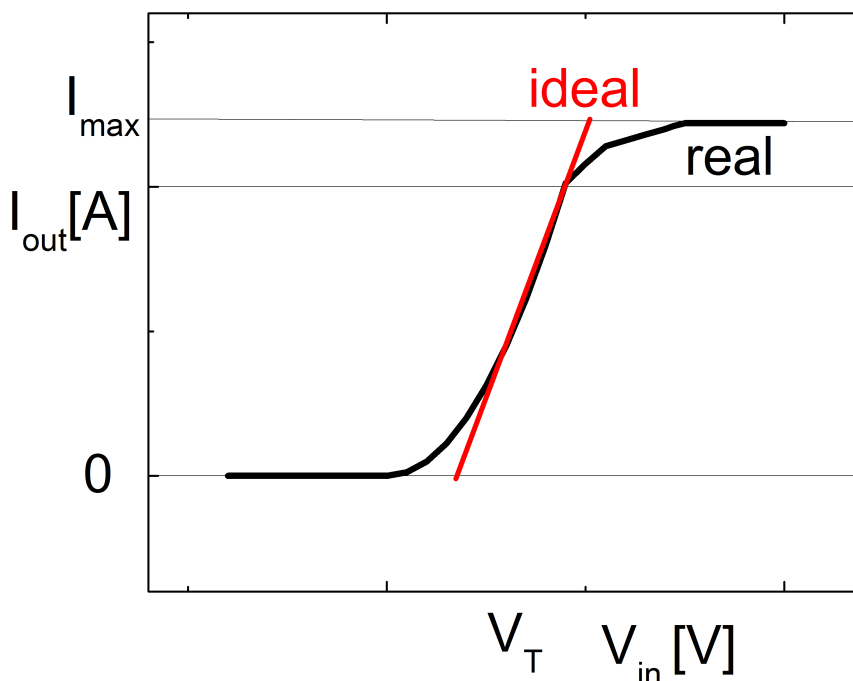
2.1 Active Modeling: Large-Signal Model of a GaAs Field Effect (Block 2: Task 8)

Educational targets

- Understand some units like dB, dBm, and dBc and some numbers.
- Understand principal ideas of large-signal approaches:
- Use large-signal models practically
- Understand power and gain compression.
- Plot dynamic voltages from the powers in relation to the a and b waves.
- Understand self-biasing for the large-signal situation in contrast to fixed current in the linear small-signal situation.
- Understand in principal current and charge equations of a large-signal model in contrast to small-signal model.
- Understand bias dependence of large-signal quantities

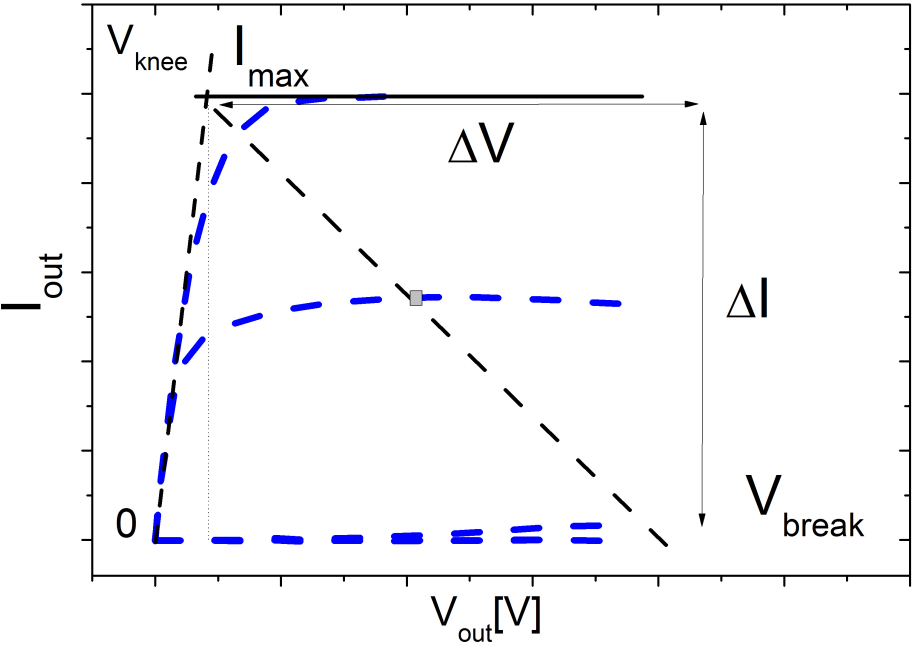
5.27

Large-Signal Situation



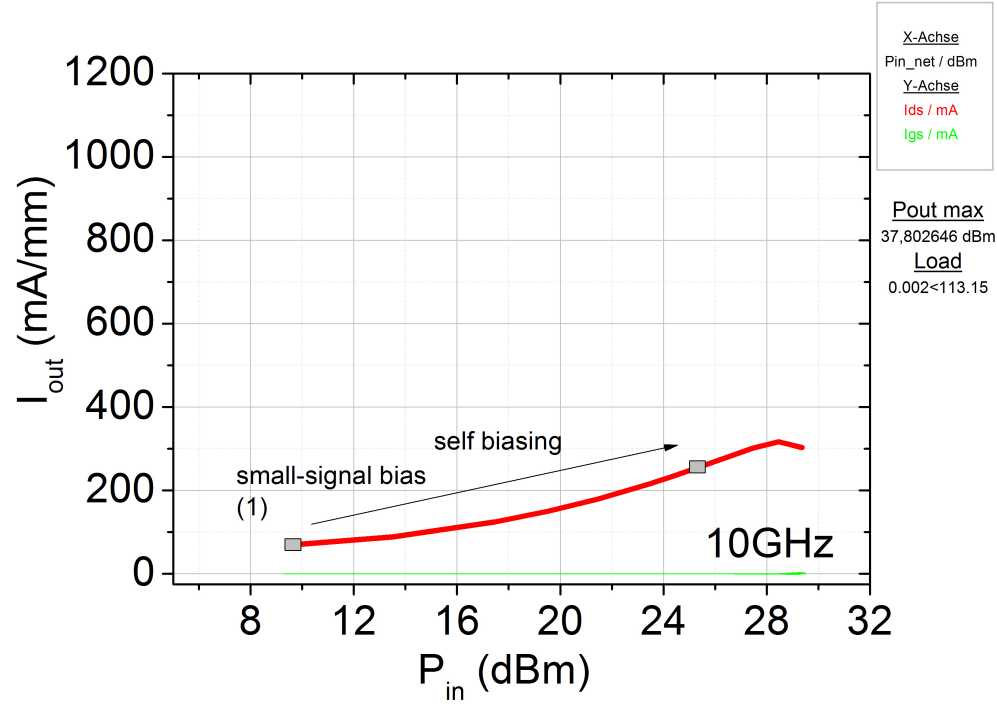
5.28

DC-Output



5.29

Large-signal



5.30

Sij Waves

a_i and b_j waves: voltage /power waves which can be visualized...

$$S_{11} = \left. \frac{b_1}{a_1} \right|_{a_2 = 0} \quad S_{22} = \left. \frac{b_2}{a_2} \right|_{a_1 = 0}$$

$$S_{21} = \left. \frac{b_2}{a_1} \right|_{a_2 = 0} \quad S_{12} = \left. \frac{b_1}{a_2} \right|_{a_1 = 0}$$



5.31

Model structure: Use Sij data over bias

Create information of Sij over bias (e.g. VGS, VDS)

Create a current equation: e.g. BJT

$$I_C = I_S \left(\exp\left(\frac{V_{BE}}{V_T}\right) - 1 \right) \quad (1)$$

Create a charge equation equation Q (and derive the capacitances C form it)

$$Q = Q(V_{DS}, V_{GS}, V_T), C = \frac{\delta Q}{\delta V} \quad (2)$$

5.32

a and b Waves

a_i and b_j waves: voltage /power waves which can be visualized...

$$a_n = \frac{1}{2} (u_n + Z_0 \cdot i_n) / \sqrt{Z_0}$$

$$b_n = \frac{1}{2} (u_n - Z_0 \cdot i_n) / \sqrt{Z_0}$$

$$u_n = (a_n + b_n) \cdot \sqrt{Z_0}$$

$$i_n = (a_n - b_n) / \sqrt{Z_0}$$

5.33

$$P_{out, netto} = |b_2|^2 - |a_2|^2$$

$$P_{in, netto} = |a_1|^2 - |b_1|^2$$

$$G = P_{out} - P_{in}$$

5.34

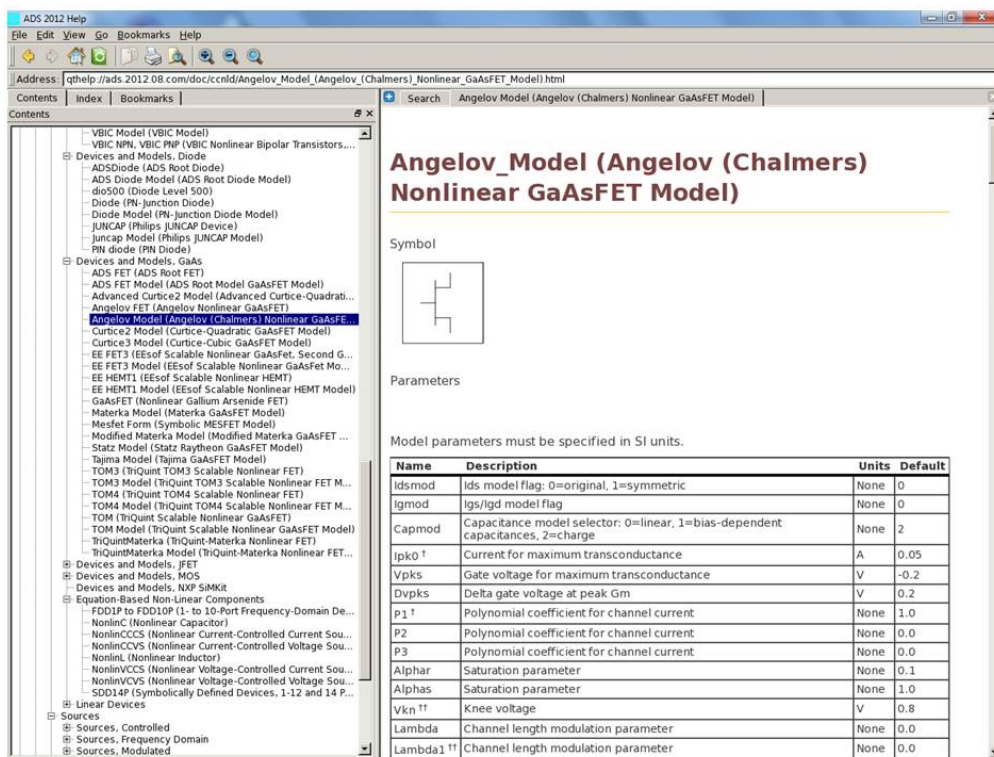
$$P \text{ in dBm} = 10 \log \left(\frac{P(mW)}{1mW} \right)$$

$$10 \log \left(\frac{1000mW}{1mW} \right) = 10 \cdot 3 = 30 \text{ dBm}$$

5.35

Large-Signal Model

Angelov FET Model



Angelov_Model (Angelov (Chalmers) Nonlinear GaAsFET Model)

Symbol



Parameters

Model parameters must be specified in SI units.

Name	Description	Units	Default
idsmod	ids model flag: 0=original, 1=symmetric	None	0
lgmod	lgs/lgd model flag	None	0
Capmod	Capacitance model selector: 0=linear, 1=bias-dependent capacitances, 2=charge	None	2
Ipk0 [†]	Current for maximum transconductance	A	0.05
Vpks	Gate voltage for maximum transconductance	V	-0.2
Dvpks	Delta gate voltage at peak Gm	V	0.2
p1 [†]	Polynomial coefficient for channel current	None	1.0
P2	Polynomial coefficient for channel current	None	0.0
P3	Polynomial coefficient for channel current	None	0.0
Alphar	Saturation parameter	None	0.1
Alphas	Saturation parameter	None	1.0
Vkn ^{††}	Knee voltage	V	0.8
Lambda	Channel length modulation parameter	None	0.0
Lambda1 ^{††}	Channel length modulation parameter	None	0.0

Example: Angelov Model for Nonlinear Devices

Current Equations

Current equations of a GaAs FET: Plot I_{ds} vs. V_{DS}

ADS 2012 Help

File Edit View Go Bookmarks Help

Address: q:\help\ads 2012 08 com\doc\cclnd\Angelov_Model_(Angelov_(Chalmers)_Nonlinear_GaAsFET_Model).html

Contents Index Bookmarks

Search Angelov Model (Angelov (Chalmers) Nonlinear GaAsFET Model)

Ids Equations

$$P1m = P1 \times (1 + B1 / \cosh^2(B2 \times Vds))$$

$$Vpkm = VPKS - DVPKS + DVPKS$$

$$\times \tanh(ALPHAS \times Vds) - VSB2 \times (Vdg - VTR)^2$$

$$\Psi = P1m \times (Vgs - Vpkm) + P2 \times (Vgs - Vpkm)^2 + P3 \times (Vgs - Vpkm)^3$$

$$\alpha = ALPHAR + ALPHAS \times (1 + \tanh(\Psi))$$

For the original model (idsmod=0)

$$Ids = IPK0 \times (1 + \tanh(\Psi)) \times \tanh(\alpha \times Vds)$$

$$\times (1 + LAMBDA \times Vds + LSB0 \times \exp(Vdg - VTR))$$

For the symmetric model (idsmod=1)

$$\Psi_n = P1m \times (Vgd - Vpkm) + P2 \times (Vgd - Vpkm)^2 + P3 \times (Vgd - Vpkm)^3$$

$$\alpha_n = ALPHAR + ALPHAS \times (1 + \tanh(\Psi_n))$$

$$\lambda_n = LAMBDA + LVG \times (1 + \tanh(\Psi_n))$$

$$\lambda_p = LAMBDA + LVG \times (1 + \tanh(\Psi))$$

$$\lambda_{n1} = LAMBDA1 + LVG \times (1 + \tanh(\Psi_n))$$

$$\lambda_{p1} = LAMBDA1 + LVG \times (1 + \tanh(\Psi))$$

$$Idsp = IPK0 \times (1 + \tanh(\Psi)) \times (1 + \tanh(\alpha \times Vds))$$

$$\times (1 + \lambda_p \times Vds + \lambda_{p1} \times \exp(\frac{Vds}{Vkn} - 1))$$

$$Idsn = IPK0 \times (1 + \tanh(\Psi_n)) \times (1 - \tanh(\alpha_n \times Vds))$$

$$\times (1 - \lambda_n \times Vds - \lambda_{n1} \times \exp(\frac{Vds}{Vkn} - 1))$$

$$Ids = 0.5 \times (Idsp - Idsn)$$

Igs, Igd Equations

5.37

2.2 Example

Example Model

ADS 2012 Help

File Edit View Go Bookmarks Help

Address: q:\help\ads 2012 08 com\doc\cclnd\Angelov_Model_(Angelov_(Chalmers)_Nonlinear_GaAsFET_Model).html

Contents Index Bookmarks

Search Angelov Model (Angelov (Chalmers) Nonlinear GaAsFET Model)

Ids Equations

$$P1m = P1 \times (1 + B1 / \cosh^2(B2 \times Vds))$$

$$Vpkm = VPKS - DVPKS + DVPKS$$

$$\times \tanh(ALPHAS \times Vds) - VSB2 \times (Vdg - VTR)^2$$

$$\Psi = P1m \times (Vgs - Vpkm) + P2 \times (Vgs - Vpkm)^2 + P3 \times (Vgs - Vpkm)^3$$

$$\alpha = ALPHAR + ALPHAS \times (1 + \tanh(\Psi))$$

For the original model (idsmod=0)

$$Ids = IPK0 \times (1 + \tanh(\Psi)) \times \tanh(\alpha \times Vds)$$

$$\times (1 + LAMBDA \times Vds + LSB0 \times \exp(Vdg - VTR))$$

For the symmetric model (idsmod=1)

$$\Psi_n = P1m \times (Vgd - Vpkm) + P2 \times (Vgd - Vpkm)^2 + P3 \times (Vgd - Vpkm)^3$$

$$\alpha_n = ALPHAR + ALPHAS \times (1 + \tanh(\Psi_n))$$

$$\lambda_n = LAMBDA + LVG \times (1 + \tanh(\Psi_n))$$

$$\lambda_p = LAMBDA + LVG \times (1 + \tanh(\Psi))$$

$$\lambda_{n1} = LAMBDA1 + LVG \times (1 + \tanh(\Psi_n))$$

$$\lambda_{p1} = LAMBDA1 + LVG \times (1 + \tanh(\Psi))$$

$$Idsp = IPK0 \times (1 + \tanh(\Psi)) \times (1 + \tanh(\alpha \times Vds))$$

$$\times (1 + \lambda_p \times Vds + \lambda_{p1} \times \exp(\frac{Vds}{Vkn} - 1))$$

$$Idsn = IPK0 \times (1 + \tanh(\Psi_n)) \times (1 - \tanh(\alpha_n \times Vds))$$

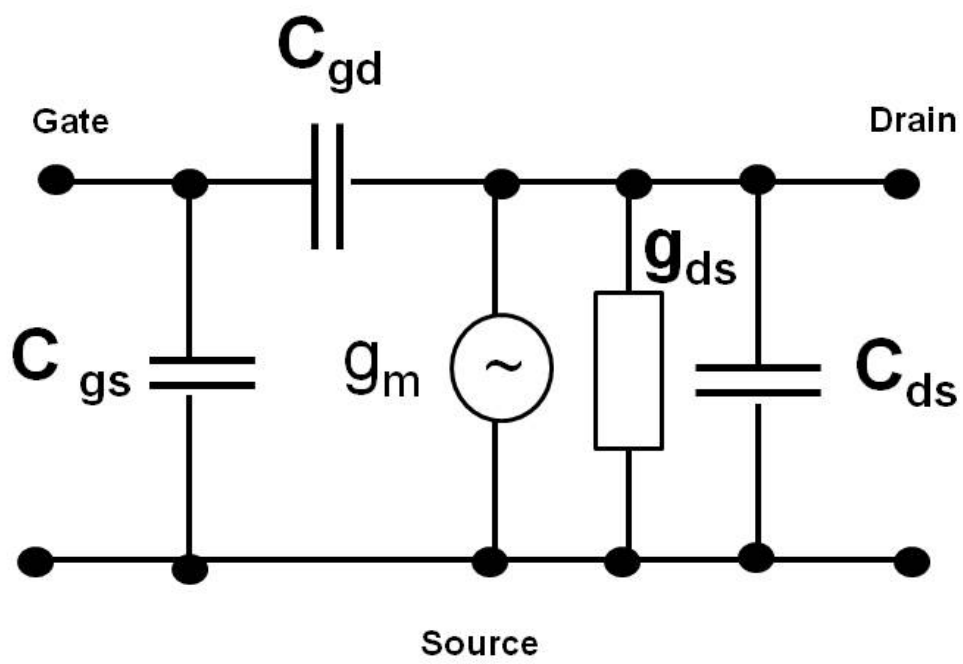
$$\times (1 - \lambda_n \times Vds - \lambda_{n1} \times \exp(\frac{Vds}{Vkn} - 1))$$

$$Ids = 0.5 \times (Idsp - Idsn)$$

Igs, Igd Equations

5.38

Equivalent Circuit FET: Intrinsic Device



3 Tasks 8 and 9: Large-Signal Models and Simulation (Block 2b)

3.1 Tasks 8 and 9: Large-Signal Models and Simulation (Block 2b)

Tasks 8

1. Explanation: Go to the example: `MWCircuits-LargeSig_Amp.wrk`. Check the top ReadMe for the Schematic.
2. In the schematic AMPPAE, perform a harmonic balance simulation at 1 GHz for $V_{ds} = 6$ V supply. Vary the input power from -30 dBm to + 8 dBm, (plot) (5 points).
3. Compare the simulation to the same performed at $V_{ds} = 12$ V and explain what is changing in gain and output power. As the supply voltage doubles, what could you expect as the maximum increase for the saturated output power ? (one plot +3 sentences) (5 points).
4. Give the power of the second harmonic (2 GHz) in dBc relative to the power at 1 GHz in dBm (i.e dB relative to the carrier, here 1 GHz.) at one voltage of 6 V (plot) (5 points).
5. Describe what happens to the PAE at $V_{ds} = 6$ V in high compression (observe Step 1 dB) (plot plus three sentences)(5 points).
6. Plot P_{out} at 1 GHz in dBm vs. P_{in} in dBm and gain in dB vs. P_{in} in dBm at 1 GHz in ONE graph for $V_{ds} = 12$ V (plot) (5 points)

5.40

Tasks 8 continued

7. Describe what the term gain compression or power compression means for this example ? (one plot) What is the compression level of the gain for an input power level of 9 dBm for 6 V(one sentence) (5 points).
8. Plot the change of the DC-current I_C in A vs. input power in dBm under RF-drive for the previous example (one plot in A). You need to search this variable ! (plot and two sentences) (5 points)
9. What happens in this example, if the supply voltage is lower than 6 V. One plot of the gain (plot). What happens ...? Three sentences max. (5 Points)

5.41

Tasks 9

1. Explanation: This part is to go to the dynamic load line (DLL) analysis to understand RF also in voltages and currents when changing loads and matching conditions.
2. Explanation: Choose the schematic DLL2.
3. Task: Explain what is the ideal positioning of the load in the diagram for this bias ? (max. 5 sentences, one plot) (5 points)
4. Task: Explain what happens with the real part of this ideal loadline if you choose a higher supply voltage: Express it in Ohms to help you... (three sentences) (5 points)

5.42

Tasks 8 and 9 format for 50 points

1. 8.2 Give plot with the right scaling... (5 point)
2. 8.3 give plot and three sentences..(5 points)
3. 8.4 Plot ! (5 points)
4. 8.5 (Plot plus three sentences)(5 points).
5. 8.6 (plot) (5 points)
6. 8.7 one plot plus one sentence (5 points)
7. 8.8 (plot and two sentences) (5 points)
8. 8.9 (plot) plus three sentences max. (5 Points)
9. 9.3 (max. 5 sentences, one plot) (5 points)
10. 9.4 Three sentences plus 5 sentences (5 points)

5.43